

Tyler Ni

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EDUCATION

New York University, Leonard N. Stern School of Business

New York, NY

B.S. in Business (Concentrations in Finance and Computing & Data Science), Computer Science

May 2025

- **Cumulative GPA:** 3.7/4.0 | **Honors:** Dean's List
- **Relevant Coursework:** Fundamentals of Machine Learning, Programming and Data Science, Statistics/Regression, Data Structures

PROFESSIONAL EXPERIENCE

C.H. Robinson

Redwood City, CA

Software Engineer II

Jul 2026 - Present

- Built an agentic pricing optimization tool using LangGraph agents, Temporal durable workflows, and Google OR-Tools CP-SAT Solver to automate computations of reasonable margin bounds for transactional rates, delivering proactive recommendations to AMs
- Designed a multi-channel proactive messaging system with configurable delivery routing between a Teams bot proxy and an enterprise agent registry, enabling alerts backed by heartbeat-based fault tolerance and LangGraph conversation checkpointing
- Architected a monorepo platform spanning FastAPI orchestration, React dashboards, and K8s-deployed Temporal workers with KEDA autoscaling, supporting concurrent workloads across 20+ pods with row-level security and Kafka-based CDC auditing

Software Engineer

Sep 2025 - Jul 2026

- Migrated Airflow data pipelines from Astronomer to self-hosted AKS, implementing Azure Workload Identity auth for Snowflake, Mongo and Redis, building reusable tools (pod operators, Vault sidecar annotation helpers) to standard secret injection across DAGs
- Architected a distributed workflow and observability framework with step-level execution, structured error classification, and Kafka-based event logging, enabling end-to-end audit trails and faster root-cause analysis across services, reducing debugging time
- Built a real-time intra-MX pricing engine that generates rates by querying historical moved-load data from Mongo using geospatial lane queries, percentile-based costing with confidence scoring and customer-specific margin strategies serving API and Streamlit

Data Science Intern

Jun 2025 - Sep 2025

- Engineered and deployed a multi-agent LLM pipeline utilizing LangGraph, Tavily, Gemini, and Azure OpenAI APIs to automate peer financial document analysis, integrating RAG with PGVector for efficient document retrieval and similarity matching
- Developed data quality validation and reporting agents that analyze macro trends, and self-evaluate output quality, iteratively improving analysis accuracy and delivering formatted financial reports, determining optimal committed to spot freight mix
- Facilitated the end-to-end design and development of a Prophet-LGBM ensemble costing model running on 2,000 clusters, achieving a \$100K weekly increase in AGP, optimizing inference latency to less than 4 seconds via model autoscaling
- Optimized prediction weighting, training on residuals to market data with the LGBM model to generate SHAP values
- Generalized the model by implementing spatial and outlier weighting and integrating Kalman filter for noise reduction
- Monitored and diagnosed data leakages within the costing model by collecting key metrics (e.g., Ridge/Lasso SHAP values) using Prometheus and visualizing insights via Datadog and interactive Streamlit dashboards

HP: Hewlett Packard

Vancouver, WA

Technical Business Analyst Intern

May 2024 - Aug 2024

- Developed and hyper-tuned Prophet time series models to forecast potential loss of HP+ Platform-as-a-Service (PaaS) units
- Leveraged Databricks SQL to develop and maintain dashboards on 10B+ rows of data that directly informed strategic investment decisions regarding the identification and rejection of non-HP ink cartridges when inserted into HP-branded printers
- Designed a full-stack authentication system on a Ras Pi with NodeJS, C ++, and Flask, deployed on AWS for IoT communication between devices. Used in sales calls to demonstrate authentication chip's capabilities, helping to convert 5+ sales opportunities
- Tested and evaluated different cryptographic protocols for authentication chips, using a combination of asymmetric and symmetric encryption. Started work on post quantum compute (Sphincs+) capabilities, while balancing memory limitations of the PCB

Openprise

San Mateo, CA

Software Engineering Intern

May 2023 - Sep 2023

- Experimented scikit CountVectorizer, TfidfTransformer, and LogisticRegression improving bot task classification accuracy by 30%
- Built a multimodal LLM customer support chatbot to processed data from pdf and video, and respond to customers questions
- Built front-end platform for chatbot with React to simplify new project creation via interface, before migrating to AWS CloudSearch.
- Defined core data needs and streamlined the delivery of cross-department data tagging and creation of standardized datasets

SKILLS & INTERESTS

- **Skills:** Python (Pandas, Numpy, Scikit, Plotly, LangGraph, Pytorch, TensorFlow), SQL, R Studio, C, C++, Java, Microsoft Excel
- **Certifications:** Machine Learning with Python (LinkedIn), Python Data Structures and Algorithms (LinkedIn)